Elizabeth (Ellie) Wynkoop

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Education

HARVARD UNIVERSITY Cambridge, MA

B.S., Applied Mathematics and Physics | GPA 3.714 *Expected Graduation May 2029*

Related Coursework: Multivariable Calculus (Math 21A); Physics and Puzzles with Cumrun Vafa (FYS23P); Applied Physics: Mechanics and Electromagnetism (AP50A-B) four engineering projects; Introduction to Probability (Stat 110) and Basics of Mechanical Engineering (ES51) to be completed and Introduction to Proofs (Math 101) by end of 2026–27.

Experience

COLLAGE AI Cambridge, MA

Product Team *June 2026 – Present*

- EdTech startup building technology for active-learning environments, in 100+ classrooms at institutions including Duke, Cornell, and Harvard – Product Team main tasks: writing product requirement documents, quality assurance
- Constructed the company Internal Knowledge Base for future hires to reference relevant documents
- Researched and identified 200+ physics professors with project-based classes to build the Collage Project Library
- Collaborated closely with Harvard Physics Professor Kelly Miller on outreach to professors, totaling 100+ pilots
- Co-built a custom team-based learning prototype using quality assurance and prompt engineering, consolidating 5+ teaching platforms into one; prototype will be used in Miller's class in Fall 2026

HARVARD JOHN A. PAULSON SCHOOL OF ENGINEERING AND APPLIED SCIENCES Cambridge, MA

Applied Physics 50 Classroom Assistant *August 2026 – December 2026*

- Grade and evaluate student presentations and homework and provide quality feedback
- Assist with 3-hour student lab sessions alongside researcher Charles Brooks
- Coach teaching fellows on the new learning platform and provide ongoing support

Projects

Optimized Windmill, 2027 Collegiate Wind Competition (in progress) **Cambridge, MA**

- Harvard Undergraduate Clean Energy Group BUILD team; work with lab heads and club upperclassmen
- Draft CAD models of wind turbine blades, brainstorm and build electric models, and discuss strategy with the research team

Rube Goldberg Machine & Precision-Calibrated Toy Car, AP50 Engineering Project Fair **Cambridge, MA**

- Earned 1st place in two project fairs against 20 other Harvard Undergraduate Teams for my team's (4 teammates) Rube Goldberg Machine and Precise Calibrated Toy Car projects
- Rube Goldberg Machine required to incorporate 7 mechanics concepts, run for 20 seconds, and ring a bell
- Precision car required to go a surprise distance the day of the fair using no technology or motor, carrying a 50g weight

Leadership & Activities

HARVARD UNDERGRADUATE ASSOCIATION **Cambridge, MA**

Elm Yard Representative *January 2026 – May 2026*

- Represent ¼ of the Class of 2029 at monthly meetings focused on student spirit and community

HARVARD PRODUCT TEAM **Cambridge, MA**

Fellow *September 2026 – Present*

- Learn the fundamentals of project management in weekly 2-hour sessions with hands-on activities led by recruited students

HARVARD GENDER INCLUSIVITY IN MATHEMATICS **Cambridge, MA**

Tech Chair *January 2026 – August 2026*

- Proctored a panel on college math research, led discussions on website design and upkeep, and helped create a welcoming environment for minority math communities

Skills & Interests

Technical: Python, C++, Java, Claude Code + Design, CAD, GitHub

Curricular: Multivariable Calculus, Linear Algebra, Proof-Based Math, Physics (Mechanics and E&M)

Interests: Painting, puzzles and board games, running, volunteering and activism